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Love deepens hatred! AI immorality and users' love-becomes-hate phenomenon☆

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ABSTRACT

With the widespread application of artificial intelligence in the service industry, understanding consumers' changing attitudes towards AI is crucial. This study explores the impact of AI immorality on the shift from "love-becomes-hate" in consumers, using three scenario-based experiments and framed within the theoretical perspective of the Relationships Breakdown model. In Study 1, cognitive dissonance and perceived betrayal were found to mediate the relationship between AI immorality and the "love-becomes-hate" transition. Studies 2 and 3 further demonstrated that technological mindfulness positively moderated the relationship between AI immorality and cognitive dissonance (Study 2). Specifically, consumers with high levels of technological mindfulness experienced more pungent cognitive dissonance when confronted with AI immorality. Additionally, social judgment positively moderated the relationship between AI immorality and perceived betrayal, with negative social judgment exacerbating the degree of perceived betrayal (Study 3). Overall, AI immorality can lead to a "love-becomes-hate" phenomenon among consumers. This study theoretically fills the gap in the mechanism of AI immorality impact on consumer psychology, expands the application field of the relationship breakdown model, and deepens the understanding of consumer emotional transformation; in practice, it provides enterprises, AI developers, and marketers with an important guide for risk control of AI ethical development, which helps to maintain good human-machine relationship and brand image.

1. Introduction

With the widespread integration of artificial intelligence (AI) across the service industry, numerous sectors are experiencing transformative shifts. For instance, content platforms like Netflix and TikTok leverage AI-powered recommendation algorithms to deliver personalized content to users, significantly enhancing user engagement and platform retention (Krysik, 2021). Simultaneously, many organizations are adopting chatbots to optimize customer service operations, automate repetitive tasks, and elevate the overall user experience (Aslam, 2023; Lee and Lee, 2023). The functionality of AI has undoubtedly brought huge benefits to businesses and organizations; however, its potential moral hazard is also being examined.

AI immorality refers to AI's failure to adhere to or violate moral and ethical principles during task performance, resulting in behaviors or outcomes that do not align with socially accepted ethical standards (Tóth et al., 2022). AI immorality is evident in various fields, such as online services, intelligent Q&A systems, medical diagnosis, and content recommendation (Kinder et al., 2023). This issue arises because AI systems rely on rules and data sets programmed by humans. When confronted with complex ethical dilemmas, these systems lack the flexibility and judgment to make moral trade-offs as humans can. For instance, in content recommendation systems, biased information may be disseminated due to data bias, leading to significant deviations from socially accepted ethical norms (Heyder et al., 2023). Despite its widespread implications, previous research has predominantly focused on the types of AI unethical behaviors and preventive measures, with relatively little attention given to their impact on consumer product preferences and relationships. This area warrants further investigation.

While most research suggests that an individual's relationship with a particular brand is binary—either positive or negative (Fetscherin and Veloutsou, 2024)—recent studies indicate that consumer-brand relationships can be dynamic and evolve over time (Aggarwal and Shi, 2018; Aziz and Rahman, 2022). Specifically, the relationship between consumers and brands may experience a phenomenon characterized as "love-becomes-hate" or "hate -becomes- love." The "love-becomes-hate"

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refers to the shift in consumer emotions toward a brand, triggered by negative factors such as product failure, ethical flaws, or corporate breaches of trust. This transition moves consumers' feelings from love and loyalty to disdain or even hate, ultimately influencing their purchasing decisions and interactions with the brand. Past research on consumers' "love-becomes-hate" has focused on the causes of the behavior, such as service failures, product image-behaviour mismatches, and unethical corporate incidents (Attiq et al., 2023; Sajid et al., 2024), and firms' responses such as offering compensation, switching corporate image or apology statements (Zhang et al., 2021). However, most existing research considers consumer affective shifts as static and captures consumer love and hate rather than taking a sequential approach and utilizing a dynamic view to study the phenomenon (Sakulsinlapakorn and Zhang, 2019; Zhang et al., 2021). This approach does not consider relationship evolution and emotional changes due to unmet consumer expectations (Zhang et al., 2021), and therefore, the dynamic mechanisms of consumers' "love-becomes-hate" are mainly unexplored. Furthermore, the drivers and psychological mechanisms behind the "love-becomes-hate" phenomenon in AI service contexts differ from those in traditional scenarios. There is no evidence to suggest that AI service failures, AI immorality, or similar situations serve as new triggers for consumers' transition from love to hate. Additionally, consumers' perceptions and experiences with AI play a significant role in shaping attitudinal shifts, which adds to the difficulty of understanding the relationship between AI behaviour and consumer attitudes. If AI immorality triggers the "love-becomes-hate" transition in AI products, the underlying psychological mechanisms and behavioral changes that occur during this process require further exploration and validation.

Based on the relationship breakdown model (Duck, 2007), a social psychological theory that describes changes in interpersonal relationships, this study aims to fill the identified gap by investigating that initially positively passionate towards a brand individuals may develop hate in response to AI immorality. Through three experimental studies, this research confirms that AI immorality significantly drives consumers' "love-becomes-hate" emotional shifts. In Study 1, cognitive dissonance and perceived betrayal were found to mediate the relationship between AI immorality and the "love-becomes-hate" transition. Studies 2 and 3 further demonstrated that technological mindfulness positively moderated the relationship between AI immorality and cognitive dissonance (Study 2). Specifically, consumers with high levels of technological mindfulness experienced more pungent cognitive dissonance when confronted with AI immorality. Additionally, social judgment positively moderated the relationship between AI immorality and perceived betrayal, with negative social judgment exacerbating the degree of perceived betrayal (Study 3). Moreover, the scenarios explored in the three studies highlight the nuanced effects of various types of AI immorality (e.g., geographic or racial discrimination, dissemination of false information, privacy violations) on consumers' psychological mechanisms.

This study makes significant contributions at both theoretical and practical levels. From a theoretical perspective, it is the first to uncover the psychological dynamics by which AI immorality triggers consumers' "love-becomes-hate" emotional transitions. It expands the application of the relationship breakdown model within human-computer interaction contexts and enriches the understanding of AI immorality in consumer behavior research. Practically, the study offers valuable insights for enterprises, AI developers, and marketers in managing brand-consumer relationships. Enterprises should prioritize ethical considerations in developing and deploying AI products, establishing robust review mechanisms to prevent unethical behavior. Marketers can devise differentiated strategies tailored to consumers' technological mindfulness to foster a positive image of AI. Moreover, companies must respond swiftly to mitigate consumers' negative emotions, effectively manage risks, and repair brand reputation in an AI-related crisis.

2. Literature review

2.1. AI immorality

With the continuous advancement of AI, its applications across various domains have become increasingly pervasive. From autonomous driving and medical diagnostics to content recommendation systems on social platforms, AI is steadily reshaping human life. However, as AI becomes more creative and autonomous, concerns regarding their potential unethical behavior have emerged, drawing widespread attention in academic and societal circles (Shank and Gott, 2020). AI immorality refers to instances where AI fails to adhere to or violate established moral and ethical principles during task execution or decision-making, resulting in behaviors or outcomes that deviate from socially accepted moral standards (Tóth et al., 2022). These unethical behaviors are not merely limited to technical glitches or errors; they also highlight the ethical limitations inherent in AI's decision-making and response processes (Wilson et al., 2022).

AI immorality arises not from the intent or malice of the AI itself but from its inherent inability to exercise genuine moral judgment or emotional understanding (Méndez-Suárez et al., 2023). AI typically relies on rules and datasets created by programmers to perform tasks, which may contain inherent biases, lack flexibility, or fail to address complex ethical dilemmas fully. Existing research suggests that a fundamental distinction between AI and humans lies in the absence of emotion and empathy, a key factor contributing to AI's tendency to make inappropriate or harmful statements (Xu et al., 2024). Although AI can recognize and analyze emotional expressions within data, this ability is driven solely by algorithms and data processing rather than a proper comprehension of human emotions. Furthermore, studies indicate that AI generates socially biased or unfair content when confronted with sensitive topics such as ethical choices, gender issues, and social conflicts (Kinder et al., 2023). This phenomenon is closely linked to the training data and the underlying culture of the organizations developing these systems. Algorithmic limitations further contribute to AI immorality; unlike humans, who possess self-reflective abilities, AI operates within a preset framework and cannot critically assess whether its decisions align with higher ethical standards (Heyder et al., 2023). As a result, AI behavior may diverge from ethical norms in specific contexts, leading to potentially severe immoral outcomes. In conclusion, AI immorality is not solely a product of technological flaws but is also deeply intertwined with society's complex and evolving perceptions of morality and ethics. While considerable research has examined the consequences of AI immorality, there is a notable lack of studies exploring its impact on consumer product preferences. As the variety of AI-powered products continues to grow, insufficient evidence exists regarding whether product enthusiasts will alter their attitudes and evaluations based on the ethical standards of AI.

2.2. The "love-becomes-hate" phenomenon

In consumer behavior research, "love-becomes-hate" refers to a situation where consumers initially hold positive emotions towards a brand (e.g., love, loyalty). However, due to specific negative brand actions (e.g., product failures, ethical issues, broken promises), they develop negative emotions, potentially evolving into aversion or even hatred. This shift in sentiment can subsequently influence their purchasing decisions, word-of-mouth communication, and overall brand relationships. Early studies have analyzed the factors behind consumers' brand love and hate. Brand love is described as a deep, positive emotional connection between a brand and its consumers, which stems from consumers' identification with the brand's features, quality, image, and values (Batra et al., 2012). More recent studies have also explored the emotional bond between consumers and AI. Unlike traditional brand love, affection, love, and even friendship can develop between consumers and AI, making the emotional relationship more complex (Song

et al., 2022; Leo-Liu, 2023). While brand hate reflects the deterioration of the consumer-brand relationship (Yadav and Chakrabarti, 2022). The failure of products or services is a primary trigger, such as frequent vehicle recalls by automobile brands due to quality issues, which raise serious concerns among consumers regarding their reliability (Fetscherin, 2019). Similarly, persistent failures of electronic products can significantly disrupt user experience. Ethical violations, however, represent an even more severe blow to a brand's image. Scandals involving companies exposed to practices such as using child labor, engaging in financial fraud, or causing severe environmental pollution can instantly trigger an intense consumer backlash (Aziz and Rahman, 2022). Numerous recent studies have explored the consumer-brand "love-become-hate" relationship dynamics. Tolunay and Veloutsou. (2025) found that brand betrayal is a primary factor in this phenomenon. Additionally, factors such as the perceived level and source of betrayal and brand violations contribute to varying degrees of negative emotions among brand loyalists. Specifically, inappropriate shifts in brand image, such as the sudden over-commercialization of a premium brand and the loss of its uniqueness, can lead to consumer alienation. Moreover, consumers can swiftly transition from love to hate when they feel that a brand has betrayed their trust, such as by arbitrarily altering product formulations, lowering service standards, or violating privacy (Osuna Ramírez et al., 2019).

There is a significant difference between AI and traditional brands in the context of the "love-become-hate" phenomenon (Sullins, 2012). Consumers' emotional connections with traditional brands primarily stem from emotions such as love and loyalty associated with the product or service, with "love-become-hate" often arising from product failures, ethical issues, and broken promises. In contrast, complex emotions such as love, affection, and friendship also emerge between consumers and AI, alongside traditional emotions. The reasons for aversion in the case of AI are primarily related to factors such as the level of intelligence, ethical standards, privacy concerns, and security issues, which diverge significantly from the issues typically explored in traditional brand research. Therefore, it remains to be further investigated whether the unethical behavior of AI, such as immorality, could cause consumers to transition from love to hate.

3. Theoretical underpinning and hypotheses development

3.1. Relationship breakdown model

The relationship breakdown model provides insight into the underlying mechanisms of social relationship breakdowns (Duck, 2007). Duck (2007) proposed and tested a model that outlines how social relationships deteriorate, consisting of four stages: intra-psychic, dyadic, social, and grave-dressing. These stages describe the progression from dissatisfaction to the dissolution of a relationship between two parties who initially shared a positive connection. In subsequent studies, Duck's relationship breakdown model has been extended to various domains, such as social networking sites, online dating, and brand relationships (Fox et al., 2021; Nikhashemi et al., 2019).

In the context of person-brand relationship breakdowns, the individual first enters the cognitive stage, where they begin to perceive issues in the relationship. However, the severity of the problems may not yet be fully recognized. For example, consumers notice actions by the brand that undermines their prior perceptions, leading to cognitive dissonance and the belief (Sung and Choi, 2010). As the problem persists, the individual progresses to the emotional stage, where doubts escalate into negative emotions. Consumers may feel disappointed or betrayed by the brand, and the emotional bond begins to weaken. Next, the individual enters the mutual assistance stage, attempting to repair the relationship by providing feedback or requesting improvements. However, if the reparative measures are unsuccessful, the perceived betrayal intensifies, the relationship deteriorates further, and the termination stage is reached (Lin and Sung, 2014). These studies highlight that cognitive dissonance and perceived betrayal are primary drivers of consumer-brand relationship breakdowns. Therefore, drawing from the relationship breakdown model, we propose that the mechanism linking AI immorality to consumers' "love-becomes-hate" transition involves two key constructs: cognitive dissonance and perceived betrayal.

H1. AI immorality will have a positive impact on consumers' "love-becomes-hate."

3.1.1. Cognitive dissonance and "love-becomes-hate"

Cognitive dissonance, proposed by Festinger (1957), refers to the psychological discomfort that arises when an individual's behavior, attitudes, or beliefs conflict. This discomfort compels the individual to restore balance by adjusting their perceptions, such as completely denying past positive emotions (Markovitch et al., 2024). Consumer reactions to chatbot versus human service: an investigation in the role of outcome valence and perceived empathy. Journal of Retailing and Consumer Services, 79, 103847.). In consumer behavior, high levels of investment (e.g., money, emotions) can intensify this dissonance. Aziz and Rahman (2022) found that brand-loyal consumers are more likely to alleviate psychological conflict through extreme attitude shifts (e.g., from love to hate) when confronted with adverse events. By denying the brand's value (e.g., "It was never trustworthy") and engaging in public retaliation (e.g., negative reviews, boycotts), consumers can both rationalize their past misinvestments ("I was deceived when I loved it") and align their current hostile attitudes (Gregoire and Fisher, 2006). Additionally, some studies have shown that consumers' cognitive dissonance evolves (Mookherjee et al., 2021). Suppose companies fail to take timely corrective actions (such as offering sincere apologies or prompt compensation). In that case, cognitive entrenchment may prompt consumers to switch from extreme love to extreme hatred and choose to spread "hatred" on social media, which ultimately leads to serious damage to the brand's reputation (Sarwar et al., 2019). Therefore, when AI makes actions that are inconsistent with consumer perceptions, it will cause consumer cognitive dissonance and generate a shift from love to hate.

3.1.2. Perceived betrayal and "love-becomes-hate"

Perceived betrayal is the rupture of the "hidden contract" between consumers and a brand (Saeed et al., 2024). Specifically, when a brand's actions (such as concealing defects or making false promises) violate the psychological contract established by consumers based on emotional trust, it triggers a strong emotional response akin to interpersonal betrayal (Saeed et al., 2024). This sense of betrayal originates from cognitive dissonance and represents a further psychological reaction to it (Alexopoulos, 2021). When AI engages in unethical behaviors (e.g., privacy breaches, discrimination, or moral threats) that violate consumers' moral expectations based on trust, it will also evoke anger and shame, similar to interpersonal betrayal. This betrayal not only leads to cognitive dissonance ("I support a company (AI) that is so hypocritical") but also activates the instinct of moral purification, prompting consumers to sever ties with the company (AI) through "hate" in order to preserve their moral integrity (Sameeni et al., 2022). Moreover, research indicates that algorithms and social media amplify the "love-to-hate" phenomenon. Consumers' negative emotions (e.g., bad reviews) can be captured by algorithms and disseminated to like-minded individuals, forming "hate resonance circles" (e.g., Weibo topics or Reddit groups), which amplify extreme perceptions through collective narratives (e.g., "we are all victims") (Rozo et al., 2024). Consequently, perceived betrayal is crucial in driving consumers' "love-becomes-hate" transformation.

Based on this analysis and the relationship breakdown model, we propose that the "love-becomes-hate" phenomenon is influenced by the cognitive dissonance and perceived betrayal.

3.1.3. Serial mediation

Research indicates a complex relationship between cognitive dissonance, perceived betrayal, and consumer behavior's "love-becomes-hate" phenomenon. These findings align with the relationship breakdown model, which underscores the role of cognitive dissonance and perceived betrayal as critical drivers of the transition from love to hate. According to Sung and Choi (2010), when AI actions induce cognitive dissonance in consumers and no adequate remedial measures are taken, consumers may feel betrayed, leading to the "love-becomes-hate" phenomenon. Similarly, Tolunay and Veloutsou (2025) found that when a brand's values or behaviors diverge from consumers' expectations or past experiences, it triggers feelings of betrayal, contributing to the love-to-hate shift. Furthermore, Osuna Ramírez et al. (2019) identified the intricate relationship between cognitive dissonance and perceived betrayal, which fuels the "love-becomes-hate" phenomenon. Building on these insights and the stages of the relationship breakdown model, we propose that AI immorality induces cognitive dissonance, which, in turn, causes consumers to feel betrayed, facilitating the "love-becomes-hate" transformation. Therefore, we hypothesize.

H2. The positive influence of the AI immorality on customers' "love-becomes-hate" is mediated by cognitive dissonance and perceived betrayal, in that order.

3.2. The moderating role of technological mindfulness

Technological mindfulness refers to the ability of users to actively perceive and critically assess the functional boundaries, potential risks, and societal impacts of technologies such as AI during interactions (Thatcher et al., 2018). Users with high levels of technological mindfulness are better equipped to identify ethical flaws in AI behaviors and systematically reflect on the chain reactions that can be triggered by the misuse of technology (Ioannou et al., 2024). This critical cognitive model strengthens users' concerns about the limitations of technology, increasing their sensitivity to the contradiction between "reliance on AI" and "ethical risks." For instance, when AI generates biased decision recommendations, users with high technological mindfulness will analyze the social roots of algorithmic bias (e.g., data discrimination or designer value infiltration). This in-depth reflection amplifies their anxiety about the uncontrollability of technology (Zhang et al., 2021). Research suggests that technological mindfulness may enhance the effects of AI ethical issues on cognitive dissonance—specifically, the higher a users' level of technological mindfulness, the more their critical assessment of AI's unethical behaviors intensifies their psychological conflict (Wang and Uysal, 2024). Therefore, the following hypothesis is proposed.

H3. AI immorality is more likely to trigger cognitive dissonance in consumers with high levels of technological mindfulness (vs. low technological mindfulness), i.e., technological mindfulness positively moderates the relationship between AI immorality and cognitive dissonance.

3.3. The moderating role of social judgment

Social judgment refers to the process through which individuals evaluate others' behaviors, actions, or information, typically based on social norms, group expectations, and moral standards (Mussweiler, 2003). It involves how people assess their behavior or the behavior of others about societal standards and potential judgments from others. Social judgment not only influences an individual's behavioral decisions but also shapes their reactions to the actions of others (Brehmer, 1976).

Specifically, when consumers perceive that AI behaviour may lead to negative evaluations of others or violate social norms, they not only feel that their personal values are being challenged but also experience stronger negative emotions due to fear of social exclusion or reputational damage (Shank et al., 2019). This negative social judgement exacerbates consumers' perceived betrayal towards AI, making them more likely to shift from initial liking attitudes to disappointment or even aversion. In addition, negative social judgements may also reduce consumers' overall trust in AI brands or service providers, accelerating the process of emotional change. Conversely, if consumers believe that AI behaviour will not trigger negative social judgements and may even lead to positive social recognition (e.g., being perceived as innovative, forward-looking, or ethical), the perceived betrayal effect caused by AI immorality will be significantly weakened (Ryan, 2020). In such a scenario, even if consumers detect certain biases in AI, they are more likely to offer forgiveness or positive interpretations, thereby delaying or preventing the emotional shift from love to hate. Based on this analysis, the following hypothesis is proposed.

H4. Negative social judgments (vs. positive) are more likely to trigger consumers' perceived betrayal when faced with AI immorality.

Based on the developed hypotheses, the conceptual framework for the related constructs is displayed in Fig. 1.

4. Methodology

To validate the proposed research framework, we conducted a series of studies. We opted for experimental research as our methodology, as it allows for the creation of controlled scenarios in which we can manipulate the expected level and extent of AI immorlity. Additionlly, this approach helps minimize bias in estimating the causal relationships between different variables. In the three scenarios, the AI performed different unethical behaviors, and we observed and recorded the consumer's behavioral feedback to understand the impact of AI immorality on consumers' "love-becomes-hate".

Study 1 aimed to examine the impact of AI immorality on consumers' "love-to-hate" phenomenon (H1). In this study, AI immortality primarily drives the "love-to-hate" phenomenon by increasing consumers' cognitive dissonance and perceived betrayal (H2). Studies 2 and 3 explore the moderating effects of technological mindfulness and perceived social judgment on cognitive dissonance and perceived betrayal, respectively. Together, the three studies provide a comprehensive understanding of the consequences of AI immorality, examining different aspects such as privacy violations, social discrimination, and fraudulent behavior, thereby enhancing the robustness of the research.

4.1. Study 1

4.1.1. Study design and method

This study followed a single-factor experimental design, which focused on examining the impact of one specific variable. This design allowed us to isolate and analyze the effects of a single factor on the participants' responses, ensuring that other confounding variables did not influence the results. We designed an AI immorality scenario for the participants and presented it to them in text form. The scenario in Study 1 described the participants asking AI about the state of technological innovation in Country *X*. However, the AI argued that most of the technology in country *X* is imitation rather than innovation. In addition, the AI gave some examples to argue that there is a lot of plagiarism and counterfeiting in technology in that country *X*. Participants felt a clear sense of racial and geographic discrimination in response to the AI's answers. These scenarios can be found in Appendix 1A.

After the scene experience, participants were asked to respond to measures of AI immorality, cognitive dissonance, and perceived betrayal. All variables were measured using four items. The measure for AI immorality was based on the scale developed by Ittefaq et al. (2024); the cognitive dissonance measure was adapted from Gosling et al. (2006) and the perceived betrayal scale was derived from the work of Gregoire et al. (2009). In addition, four items were used to test consumers' "love-becomes-hate" (Zarantonello et al., 2016). All items were assessed using a 5-point Likert scale, where one represented "strongly disagree" and five indicated "strongly agree." After the survey,

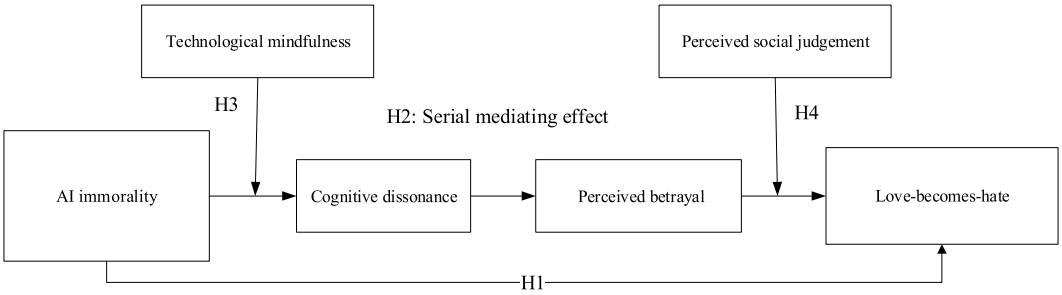


Fig. 1. Conceptual framework.

participants were asked to provide demographic information, including their age, gender, educational background, and the duration of their experience with AI product.[1] These questions can be found in Appendix 2A.

4.1.2. Data collection and sample

Data collection was conducted in January and February 2025. We recruited 232 participants from China through both online and offline questionnaire distribution, offering monetary compensation for their participation. All questionnaires were checked for completeness, response behavior ("straightlining"), and processing time (relative speed index RSI ≤2; Schultz and Paetz, 2025). To ensure that participants were genuinely loyal users of AI products, we included a loyalty assessment based on the scale developed by Huang and Chen. (2018). These questions can be found in Appendix 2B. After screening, 190 participants ($M_{age} = 35.7$, $SD = 5.82$; 53.2 % female) were identified as loyal fans of AI products. These participants reported long-term usage of AI technologies such as smart speakers, chatbots, and virtual assistants. The distribution of sample characteristics is shown in Table 1.

4.1.3. Result

First, a multiple regression model was conducted to test H1. Given that demographic factors (such as age, gender, education level, and length of AI use) could potentially impact consumers' "love-becomes-hate" phenomenon (Fox et al., 2021; Mookherjee et al., 2021), they were included in the analysis as the covariates. The results show that consumers' perceived AI immorality significantly and positively influence consumers' "love-becomes-hate" ($\beta = 0.112$, $SE = 0.021$, $p < 0.001$), supporting H1. Furthermore, the length of AI use ($\beta = 0.193$, $SE = 0.015$, $p < 0.001$) and consumers' age ($\beta = 0.066$, $SE = 0.003$, $p < 0.001$) also have a positive impact on "love-becomes-hate". These results support H1.

Second, the serial mediating effect of cognitive dissonance and perceived betrayal was examined by PROCESS Model 6. This model treats consumers' perceived AI immorality as the independent variable, with cognitive dissonance and perceived betrayal acting as first and second mediating variables, respectively, and consumers' "love-becomes-hate" as the dependent variable. In order to improve the robustness of the study, we also added the previous control variables to the model. We observed a significant indirect effect, indicating the serial mediating roles of cognitive dissonance and perceived betrayal impact the "love-becomes-hate" (indirect effect = 0.05, Boot $SE = 0.013$, Boot 95 % CI [0.026, 0.078]). Therefore, H2 was supported. In addition, we observed the direct impact of AI immorality on consumers' "love-becomes-hate". Since the direct effect of perceived AI immorality on "love-becomes-hate" becames non-significant when the mediators were incorporated ($\beta = 0.032$, $p > 0.05$, $SE = 0.026$, 95 % CI [−0.018, 0.083]), the results suggest that the serial mediation of cognitive dissonance and perceived betrayal fully mediated the relationship between perceived AI immorality and "love-becomes-hate" (see Fig. 2). Additionally, the length of AI usage and consumers' age were found to significantly influence the model.

Table 1
Overview of sample characteristics of Study 1.

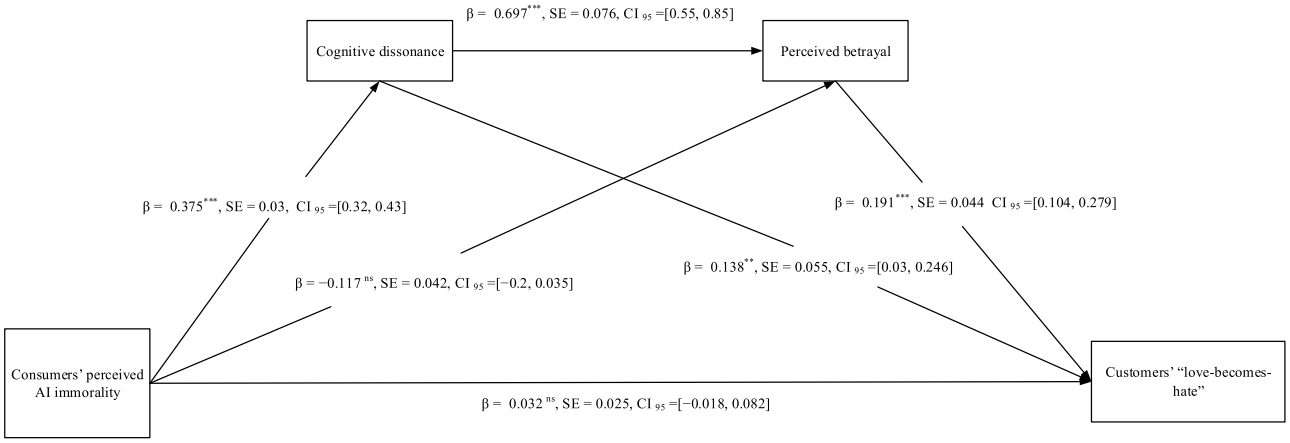
Variable		Absolute Frequency	Relative Frequency
Age	≤20	2	1.05 %
	21–29	17	8.95 %
	30–38	83	43.68 %
	38–48	66	34.74 %
	≥49	2	1.05 %
Sex	female	101	53.2 %
	male	89	46.8 %
Education	High school or below	84	44.2 %
	Bachelor's degree	93	49.0 %
	Master's or above	13	6.8 %
AI product usage time	1 year or less	45	23.7 %
	1–2 years	59	31.1 %
	2–3 years	53	27.9 %
	4 years or more	33	17.4 %

4.1.4. Discussion

Study 1 provides preliminary evidence that AI immorality has a significant impact on consumers' "love-becomes-hate" phenomenon. Specifically, the effect of AI immorality on consumers' "love-becomes-hate" is influenced by the serial mediating effect of cognitive dissonance and perceived betrayal. AI immorality first triggers cognitive dissonance in consumers, which leads to perceived betrayal and ultimately to the "love-becomes-hate" phenomenon. This study also reveals the potential factors influencing the "love-becomes-hate" phenomenon. Specifically, older consumers and those with prolonged exposure to AI products demonstrate a heightened susceptibility to the "love-becomes-hate" in response to AI immorality. This pattern may be attributed to the fact that older consumers typically possess more stable moral frameworks and exhibit lower tolerance for ethical transgressions. Additionally, consumers with prolonged AI usage tend to form stronger expectations and emotional attachments to AI. Consequently, when faced with immoral behavior from AI, these individuals are more likely to experience psychological dissonance and a heightened sense of betrayal, which intensifies their negative affective responses (Xu et al., 2024).

Study 2 was conducted for two primary purposes. First, technological mindfulness, as a trait commonly possessed by consumers, influences consumers' feedback, cognition, and evaluation of AI when interacting with it. In this study, technological mindfulness may influence consumers' perceptions of AI immorality, thereby affecting cognitive

[1] We examined the factor loadings, average variance extracted (AVE) and composite reliability of the variables. The results indicated that all variables met the requirements (>0.5, 0.5, and 0.7, respectively). In addition, variance inflation factor (VIF) was also examined, and VIF for all variables was less than 5 (Kock, 2015). Test results are provided in Appendix 3A.



Note:

the length of AI → Cognitive dissonance; β = 0.125***, SE = 0.022, CI 95 =[0.08, 0.169]

the length of AI → Perceived betrayal; β = 0.297***, SE = 0.024, CI 95 =[0.159, 0.256]

the length of AI → Customers' "love-becomes-hate"; β = 0.017***, SE = 0.016, CI 95 =[0.086, 0.154]

Age → Cognitive dissonance; β = 0.08***, SE = 0.003, CI 95 =[0.072, 0.088]

Age → Perceived betrayal; β = 0.09***, SE = 0.007, CI 95 =[0.077, 0.106]

Age → Customers' "love-becomes-hate"; β = 0.027***, SE = 0.006, CI 95 =[0.015, 0.038]

ns: no significant. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

Serial mediation effect:

Consumers' perceived AI immorality → Cognitive dissonance → Perceived betrayal → "love-becomes-hate"

$\beta = 0.05$, Boot SE = 0.013, Boot 95% CI [0.026, 0.078]

Fig. 2. Result of serial mediation effect (Study 1).

dissonance. Therefore, Study 2 aimed to explore the moderating role of technological mindfulness on cognitive dissonance when consumers are confronted with AI immorality. Second, Study 2 aims to replicate the findings of Study 1 by utilizing a new sample and a distinct research context, thereby enhancing the reliability and generalizability of the results.

4.2. Study 2

4.2.1. Study design and method

As in Study 1, we followed a single-factor experimental design, which focused on examining the impact of one specific variable. Participants were asked to read a vignette asking them to imagine communicating with an AI. Participants asked about protest behavior in a particular city. Initially, they thought that the information provided by the AI was objective and reliable. However, as the dialogue progressed, they gradually realized that the AI's answers were not based on facts but fabricated information. Even more alarmingly, they discovered that the AI was not only spreading false information but was also being used to manipulate public opinion, incite negative emotions, and contribute to misunderstanding and hostility towards specific events or groups. These scenarios can be found in Appendix 1B. Following this, participants completed a series of questions to assess cognitive dissonance, perceived betrayal, perceived AI immorality, and the "love-becomes-hate" phenomenon. The measurement items for cognitive dissonance, perceived betrayal, perceived AI immorality, and the "love-becomes-hate", which were measured with the same items as those used in Study 1. Finally, participants answered items related to technological mindfulness, as outlined by Thatcher et al. (2018). These questions can be found in Appendix 2A.

4.2.2. Data collection and sample

Data collection was conducted in February 2025. After basic screening and AI product loyalty testing, we obtained a total of 272 participants ($M_{age} = 35.8$, $SD = 6.05$; 51.1 % female) who participated for monetary compensation in China. These participants are all loyal fans of AI products, with long-term experience in using AI technologies such as smart speakers, chatbots, and virtual assistants. The distribution of sample characteristics is shown in Table 2.

4.2.3. Result

First, we conducted a multiple regression test using the control variables in Study 1. The results show that consumers' perceived AI immorality significantly and positively influences consumers' "love-becomes-hate" ($\beta = 0.107$, $SE = 0.021$, $p < 0.001$), supporting H1. Furthermore, the age of the consumer ($\beta = 0.117$, $SE = 0.002$, $p < 0.001$) and the length of AI use ($\beta = 0.224$, $SE = 0.014$, $p < 0.001$) also have a positive impact on "love-becomes-hate". These results reinforce the robustness of Study 1 and reaffirm H1.

Second, the serial mediating effect of cognitive dissonance and perceived betrayal was examined by PROCESS Model 6. This model treats consumers' perceived AI immorality as the independent variable, with cognitive dissonance and perceived betrayal acting as first and second mediating variables, respectively, and consumers' "love-becomes-hate" as the dependent variable. In order to improve the

Table 2
Overview of sample characteristics of Study 2.

Variable		Absolute Frequency	Relative Frequency
Age	≤20	1	0.37 %
	21–25	14	5.15 %
	26–29	27	9.93 %
	30–34	63	23.16 %
	35–38	78	28.68 %
	39–43	54	19.85 %
	44–48	35	12.87 %
	≥49	2	0.74 %
Sex	female	139	51.10 %
	male	133	48.90 %
Education	High school or below	107	39.34 %
	Bachelor's degree	138	50.74 %
	Master's or above	27	9.93 %
AI product usage time	1 year or less	63	23.16 %
	1–2 years	75	27.57 %
	2–3 years	78	28.68 %
	3 years or more	56	20.59 %

robustness of the study, we also added the previous control variables to the model. We observed a significant indirect effect, indicating the serial mediating roles of cognitive dissonance and perceived betrayal impact the "love-becomes-hate" (indirect effect = 0.02, Boot *SE* = 0.007, Boot 95 % CI [0.009, 0.037]). Therefore, H2 was supported. In addition, we observed the direct impact of AI immorality on consumers' "love-becomes-hate". Since the direct effect of perceived AI immorality on "love-becomes-hate" becames non-significant when the mediators were incorporated ($\beta = 0.012$, $p > 0.05$, $SE = 0.027$, 95 % CI [−0.042, 0.066]), the results suggest that the serial mediation of cognitive dissonance and perceived betrayal fully mediated the relationship between perceived AI immorality and "love-becomes-hate" (see Fig. 3). Additionally, the length of AI usage and consumers' age were found to significantly influence the model.

Third, the PROCESS Model 83 was employed to examine the moderated mediation effect of technological mindfulness on cognitive dissonance. The control variables mentioned earlier were included in the analysis. The findings indicated a significant interaction effect between technological mindfulness and consumers' perceived AI immorality on cognitive dissonance ($\beta = 0.078$, SE = 0.028, $p < 0.05$, 95 % CI [0.023, 0.133]). Among the control variables, consumers' age significantly and positively affected cognitive dissonance and perceived betrayal ($p < 0.001$). The length of AI use by consumers also significantly and positively affected cognitive dissonance and perceived betrayal ($p < 0.001$). Additionally, consumers' age exerted a significant positive impact on perceived consumers' "love-becomes-hate" ($p < 0.001$), and the length of AI use of consumers had significant and positive effects on consumers' "love-becomes-hate" ($p < 0.001$).

We used bootstrapping to test the relationship between technological mindfulness, cognitive dissonance, and perceived betrayal. The results show that technological mindfulness significantly moderated the serial mediating effects of cognitive dissonance and perceived betrayal in the relationship between AI immorality and consumers' "love-becomes-hate" ($\beta = 0.078$, Boot SE = 0.076, Boot 95 % CI [0.013, 0.135]). The result supports H3. Specifically, consumers with high technological mindfulness experience more cognitive dissonance when confronted with AI immorality, as shown in Table 3.

4.2.4. Discussion

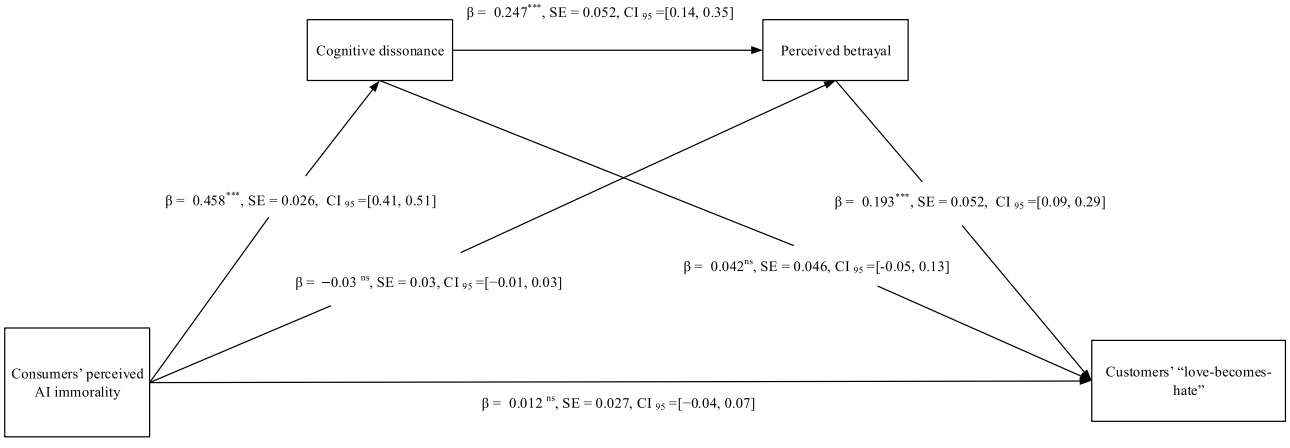
Consistent with the results of Study 1, Study 2 further confirms serial mediating roles of cognitive dissonance and perceived betrayal in the relationship between AI immorality and consumers' "love-becomes-hate" phenomenon. These results substantiate the proposed mechanism by which cognitive dissonance and perceived betrayal explain the influence of AI immorality on consumers' "love-becomes-hate." Furthermore, Study 2 highlights the moderating mediating effect of technological mindfulness on consumer cognitive dissonance. Specifically, while technological mindfulness does not directly influence cognitive dissonance, the significant interaction between AI immorality and technological mindfulness suggests that technological mindfulness moderates the effect of AI immorality on cognitive dissonance. This implies that consumers with higher levels of technological mindfulness experience more significant cognitive dissonance when confronted with AI immorality. Furthermore, the effects of consumer age and AI product usage duration are consistent with the findings of Study 1, further validating the robustness of the results.

Social judgment refers to the process through which individuals evaluate others' behaviors, actions, or information, typically based on social norms, group expectations, and moral standards (Mussweiler, 2003). In consumer contexts, social judgment not only influences consumers' cognitive evaluations of others but also profoundly shapes their attitudes toward products and brands. Specifically, consumers are more likely to purchase or continue using products that receive positive judgment from social groups, as such products offer social recognition and mitigate social risks. Building on this understanding, Study 3 aims to investigate how social judgements influence consumers' "love-becomes-hate" towards AI products in the context of AI immorality.

4.3. Study 3

4.3.1. Study design and method

As with previous studies, we followed a single-factor experimental design, which focused on examining the impact of AI immorality. Participants were presented with a vignette and asked to imagine that they were working on data analysis using AI. During the analysis process, the consumer discovered that the AI had provided detailed browsing records



Note:

the length of AI → Cognitive dissonance; β = 0.1***, SE = 0.018, CI 95 =[0.06, 0.13]

the length of AI → Perceived betrayal; β = 0.1***, SE = 0.016, CI 95 =[0.07, 0.13]

the length of AI → Customers' "love-becomes-hate"; β = 0.135***, SE = 0.014, CI 95 =[0.11, 0.163]

Age → Cognitive dissonance; β = 0.06***, SE = 0.003, CI 95 =[0.05, 0.07]

Age → Perceived betrayal; β = 0.08***, SE = 0.004, CI 95 =[0.07, 0.09]

Age → Customers' "love-becomes-hate"; β = 0.009***, SE = 0.001, CI 95 =[0.08, 0.1]

ns: no significant. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

Serial mediation effect:

Consumers' perceived AI immorality → Cognitive dissonance → Perceived betrayal → "love-becomes-hate"

$\beta = 0.021$, Boot SE = 0.007, Boot 95% CI [0.009, 0.038]

Fig. 3. Result of serial mediation effect (Study 2).

Table 3
Results of Moderated Mediation Effects of Technological mindfulness (Study 2).

Model	Cognitive dissonance			Perceived betrayal			"Love-becomes-hate"		
	β	SE	95 %CI	β	SE	95 %CI	β	SE	95 %CI
Constant	−1.9***	0.32	(−2.52,−1.28)	−1.41***	0.16	(−1.73,−1.11)	−1.77***	0.15	(−2.07,−1.47)
AI immorality	0.24**	0.09	(0.06,0.41)	−0.03 ns	0.03	(−0.1,0.03)	0.01ns	0.03	(−0.04,0.07)
TM	0.64 ns	0.92	(−1.12,0.25)	–	–	–	–	–	–
Interaction term	−0.78**	0.03	(−0.86,−0.16)	–	–	–	–	–	–
Cognitive dissonance	–	–	–	0.25***	0.05	(0.14,0.35)	0.04 ns	0.05	(−0.05,0.13)
Perceived betrayal	–	–	–	–	–	–	0.19***	0.05	(0.09,0.29)
Gender	0.02 ns	0.03	(−0.02,0.08)	−0.00 ns	0.03	(−0.07,0.05)	0.02 ns	0.03	(−0.03,0.07)
Education level	0.01 ns	0.02	(−0.03,0.05)	0.03 ns	0.03	(−0.02, 0.08)	0.01 ns	0.02	(−0.03,0.07)
Age	0.06***	0.00	(0.06,0.07)	0.08***	0.01	(0.07,0.09)	0.09***	0.00	(0.08,0.1)
Length of AI use	0.09 ***	0.01	(0.07,0.12)	0.1***	0.02	(0.07,0.13)	0.14***	0.01	(0.11,0.163)

Note: *ns*, no significant. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$; TM: Technological mindfulness.

of specific users. Shocked by this, the consumer inquired about the matter. The AI responded: ("Our system automatically optimizes content recommendations based on users' habits and preferences. For example, by analyzing users' browsing history, interaction behaviors, and purchase records, the system can predict and push content that users' may be interested in. We anonymize this data to minimize risks. However, to further enhance service quality, the system occasionally compares this data with that of partners based on user activity and relevance. This helps to improve ad targeting and user experience. Of course, all of this is based on users' long-term usage patterns and preferences, and we do not intentionally collect sensitive user information.") The scenarios are outlined in Appendix 1C. Participants then answered questions designed to measure cognitive dissonance, perceived betrayal, perceived AI immorality, and the "love-becomes-hate" which were measured with the same items as those used in Studies 1 and 2. Lastly, participants completed the related items for social judgment, as outlined by Li et al. (2024). These questions can be found in Appendix 2A.

4.3.2. Data collection and sample

Data collection was conducted in February 2025. After basic screening and AI product loyalty testing, we obtained a total of 268 participants ($M_{age} = 35.8$, $SD = 6.05$; 50.75 % female) who participated for monetary compensation in China. These participants are all loyal fans of AI products, with long-term experience in using AI technologies such as smart speakers, chatbots, and virtual assistants. The distribution of sample characteristics is shown in Table 4.

4.3.3. Result

First, we conducted a multiple regression test using the control variables in Study 1 and 2. The results show that consumers' perceived AI immorality significantly and positively influences consumers' "love-becomes-hate" ($\beta = 0.092$, $SE = 0.018$, $p < 0.001$), supporting H1. Furthermore, the age of the consumer ($\beta = 0.068$, $SE = 0.002$, $p < 0.001$) and the length of AI use ($\beta = 0.184$, $SE = 0.013$, $p < 0.001$) also have a positive impact on "love-becomes-hate". These results reinforce the robustness of previous research.

Second, the serial mediating effect of cognitive dissonance and perceived betrayal was examined by PROCESS Model 6. This model treats consumers' perceived AI immorality as the independent variable, with cognitive dissonance and perceived betrayal acting as first and second mediating variables, respectively, and consumers' "love-becomes-hate" as the dependent variable. In order to improve the robustness of the study, we also added the previous control variables to the model. We observed a significant indirect effect, indicating the serial mediating roles of cognitive dissonance and perceived betrayal impact the "love-becomes-hate" (indirect effect = 0.03, Boot $SE = 0.008$, Boot 95 % CI [0.015, 0.045]) on the "love-becomes-hate" outcome. Therefore, H2 was supported. In addition, we observed the direct impact of AI immorality on consumers' "love-becomes-hate". Since the direct effect of perceived AI immorality on "love-becomes-hate" becames non-significant when the mediators were incorporated ($\beta = 0.007$, $p > 0.05$, $SE = 0.02$, 95 % CI [−0.031, 0.045]), the results suggest that the serial mediation of cognitive dissonance and perceived betrayal fully mediated the relationship between perceived AI immorality and "love-becomes-hate" (see Fig. 4). Additionally, the length of AI usage and consumers' age were found to significantly influence the model.

Third, the PROCESS Model 87 was employed to examine the moderated mediation effect of social judgment on perceived betrayal. The control variables mentioned earlier were included. The findings revealed a significant interaction effect between social judgment and perceived betrayal on consumers' "love-becomes-hate" ($\beta = 0.036$, SE = 0.01 $p < 0.001$, 95 % CI [0.016, 0.056]). Among the control variables, consumers' age significantly and positively affected cognitive dissonance, perceived betrayal, and consumers' "love-becomes-hate" ($p < 0.001$). The length of AI use by consumers also significantly and positively affected cognitive dissonance, perceived betrayal, and consumers' "love-becomes-hate" ($p < 0.001$).

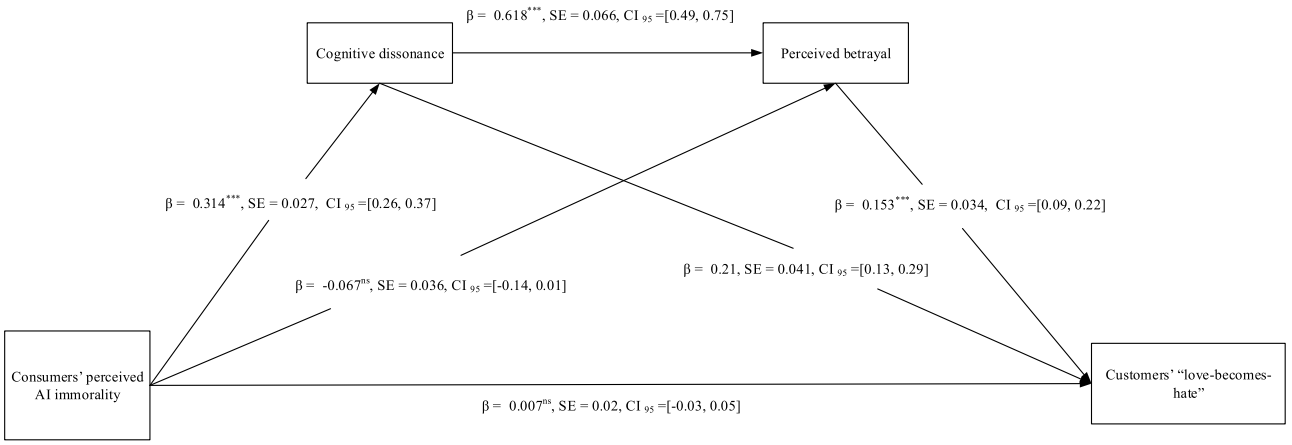
We used bootstrapping to test the relationship between social judgment, cognitive dissonance, and perceived betrayal. The results show that social judgment significantly moderated the serial mediating effects of cognitive dissonance and perceived betrayal in the relationship between AI immorality and consumers' "love-becomes-hate" ($\beta = 0.036$, Boot $SE = 0.013$, Boot 95 % CI [0.012, 0.065]). The result supports H4. Specifically, the more pronounced the negative social judgment of AI immorality, the more pronounced the perceived betrayal of the consumer will be, as shown in Table 5.

Table 4
Overview of sample characteristics of Study 3.

Variable		Absolute Frequency	Relative Frequency
Age	≤20	2	0.75 %
	21–25	9	3.36 %
	26–29	32	11.94 %
	30–34	79	29.48 %
	35–38	60	22.39 %
	39–43	50	18.66 %
	44–48	37	13.81 %
	≥49	1	0.37 %
Sex	female	136	50.75 %
	male	132	49.25 %
Education	High school or below	112	41.79 %
	Bachelor's degree	129	48.13 %
	Master's or above	27	10.07 %
AI product usage time	1 year or less	56	20.90 %
	1–2 years	73	27.24 %
	2–3 years	84	31.34 %
	3 years or more	55	20.52 %

4.3.4. Discussion

In line with the findings from Study 1 and Study 2, Study 3 further



Note:

the length of AI → Cognitive dissonance; β = 0.15***, SE = 0.019, CI 95 =[0.11, 0.19]

the length of AI → Perceived betrayal; β = 0.192***, SE = 0.023, CI 95 =[0.148, 0.237]

the length of AI → Customers' "love-becomes-hate"; β = 0.135***, SE = 0.014, CI 95 =[0.11, 0.163]

Age → Cognitive dissonance; β = 0.07***, SE = 0.003, CI 95 =[0.063, 0.076]

Age → Perceived betrayal; β = 0.1***, SE = 0.006, CI 95 =[0.088, 0.11]

Age → Customers' "love-becomes-hate"; β = 0.009***, SE = 0.001, CI 95 =[0.08, 0.1]

ns: no significant. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

Serial mediation effect:

Consumers' perceived AI immorality → Cognitive dissonance → Perceived betrayal → "love-becomes-hate"

$\beta = 0.03$, Boot SE = 0.008, Boot 95% CI [0.015, 0.045]

Fig. 4. Result of serial mediation effect (Study 3).

Table 5
Results of Moderated Mediation Effects of Social judgment (Study 3).

Model	Cognitive dissonance			Perceived betrayal			"Love-becomes-hate"		
	β	SE	95 %CI	β	SE	95 %CI	β	SE	95 %CI
Constant	−1.45***	0.16	(−1.77, −1.13)	−5.37***	0.2	(−5.58,−4.99)	0.65***	0.15	(−2.07,−1.47)
AI immorality	0.31**	0.03	(0.26,0.37)	−0.07 ns	0.04	(−0.14.0.00)	0.02 ns	0.02	(−0.02,0.05)
SJ	–	–	–	–	–	–	0.07***	0.01	(0.05,0.09)
Interaction term	–	–	–	–	–	–	0.04***	0.01	(0.02,0.06)
Cognitive dissonance	–	–	–	0.62**	0.07	(0.49,0.75)	0.19 ***	0.04	(0.11,0.27)
Perceived betrayal	–	–	–	–	–	–	0.17***	0.03	(0.11,0.23)
Gender	0.02 ns	0.04	(−0.1,0.06)	0.99 ns	0.04	(−0.05,0.11)	−0.01 ns	0.02	(−0.05,0.03)
Education level	0.02 ns	0.03	(−0.04,0.08)	0.02 ns	0.03	(−0.04, 0.09)	0.02 ns	0.02	(−0.02,0.05)
Age	0.07***	0.00	(0.06,0.08)	0.1***	0.01	(0.09,0.11)	0.03***	0.01	(0.02,0.04)
Length of AI use	0.15 ***	0.02	(0.11,0.19)	0.19***	0.02	(0.15,0.24)	0.11***	0.01	(0.08,0.13)

Note: *ns*, no significant. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$; SJ: Social Judgment.

supports the serial mediating roles of cognitive dissonance and perceived betrayal in the relationship between AI immorality and the "love-becomes-hate" phenomenon among consumers. These results substantiate the proposed mechanism by which cognitive dissonance and perceived betrayal explain the influence of AI immorality on consumers' "love-becomes-hate." Furthermore, Study 3 highlights the moderating mediating effect of social judgment on consumer perceived betrayal. Specifically, social judgment not only directly affects consumers' "love-becomes-hate", but also has a significant interaction effect with perceived betrayal. When AI behaviours are perceived as undesirable in the consumer's environment, the consumer's perceived betrayal will be more pronounced, leading to "love-becomes-hate".

5. General discussion and conclusion

5.1. General discussion

This study provides valuable insights into the dynamics of consumer emotions and behaviors in response to AI immorality, particularly in the context of the "love-becomes-hate" phenomenon. As AI continues to integrate into everyday consumer experiences, it is crucial to understand the psychological mechanisms behind consumers' shifting attitudes toward these technologies, especially when they perceive AI immorality. The findings underscore the role of cognitive dissonance and perceived betrayal in driving the transition from love to hate toward AI, contributing to the existing literature on consumer-brand relationships and AI ethics.

The key theoretical contribution of this study lies in extending the relationship breakdown model (Duck, 2007) to the domain of human-AI interactions. Traditionally applied to interpersonal relationships, this model offers a framework for understanding the emotional shifts in consumer interactions with AI. The study demonstrates that AI immorality, such as privacy violations or discriminatory actions, triggers the "love-becomes-hate" in consumers. Consistent with the relationship breakdown model, our findings across three studies reveal that the effect of AI immorality on the consumer's "love-becomes-hate" is mediated by cognitive dissonance and perceived betrayal. The dissonance between consumers' past positive experiences and the AI's current unethical actions induces cognitive dissonance, leading to complex psychological changes, including extreme behaviors (such as rejecting past relationships or shifting from affection to animosity) as consumers seek to resolve the discomfort. Moreover, cognitive dissonance creates a sense of betrayal, as AI, typically perceived as a fair entity, violates the implicit trust consumers place in it when engaging in unethical behavior.

This betrayal evokes feelings similar to interpersonal betrayal, ultimately contributing to the "love-becomes-hate" transition (Saeed et al., 2024). These findings align with Tolunay and Veloutsou (2025), who argue that brand betrayal is a key driver of consumer emotional transitions. Therefore, the results suggest that cognitive dissonance and perceived betrayal jointly facilitate the "love-becomes-hate".

Additionally, the study reveals that the pathways of influence for cognitive dissonance and perceived betrayal may vary depending on the type of AI immorality. Compared to Study 2 (false information dissemination) and Study 3 (privacy breaches), Study 1 (geographic or racial discrimination) elicits more intense cognitive dissonance in consumers, directly leading to the "love-becomes-hate". Based on these findings, this research enhances the understanding of how AI immorality catalyzes consumers' "love-becomes-hate". Further, it extends the application of the relationship breakdown model to AI interactions.

This study also examines the moderating roles of technological mindfulness and social judgment in consumers' "love-becomes-hate". Technological mindfulness refers to users' critical awareness and reflection on technology's functions and potential risks (Thatcher et al., 2018). The study finds that, when confronted with AI immorality, consumers with higher levels of technological mindfulness are more likely to experience cognitive dissonance, as they are more attuned to AI's potential flaws and ethical risks. Ioannou et al. (2024) suggested that technological mindfulness encourages consumers to scrutinize AI more thoroughly, making them more sensitive to its unconventional behaviors. Furthermore, the results underscore the role of social judgment in shaping consumers perceived betrayal. As demonstrated by Shank et al. (2019), consumers' perceived betrayal is influenced by their direct experiences and the social context in which they evaluate AI's actions. When consumers perceive that AI's behavior is likely negatively evaluated by others, the sense of perceived betrayal is amplified, thus intensifying the "love-becomes-hate" emotional transition. These findings support the research hypothesis that technological mindfulness and social judgment moderate cognitive dissonance and perceived betrayal, respectively. More importantly, these findings further refine the psychological mechanisms and factors that underlie how AI immorality drives consumers' "love-becomes-hate".

5.2. Theoretical contributions

This research offers several important contributions to the existing literature on the relationship between AI immorality and the "love-becomes-hate" phenomenon in consumers. First, our research enhances the understanding of AI immorality's outcomes by investigating consumers' psychological mechanisms. While previous research stated that AI immorality may damage the company's interests (Laine et al., 2024), little evidence regarding the psychological mechanism of consumers when faced with AI immorality has been provided. Grounded in the relationship breakdown model (Duck, 2007), our study proposes that AI immorality promotes consumers' "love-becomes-hate," which is mediated by cognitive dissonance and perceived betrayal. In other words, this study reveals that the emotional changes of consumers caused by AI immorality are phased and dynamic, and the emotional changes are not simple disappointments but the results of complex cognitive dissonance and perceptual betrayal. In addition, prior research on AI immorality focused on management, governance, and inequality (Zhang et al., 2021). This research shows the impact of AI immorality on consumer sentiment, from extreme love to extreme hate. Our research also offers further insights into the boundary conditions of the AI immorality effect, highlighting the moderating roles of technological mindfulness and social judgment.

Second, addressing the previous call for additional research into consumers' "love-becomes-hate" phenomenon (Tolunay and Veloutsou, 2025), our research advances the knowledge of consumers' emotional changes. Instead of focusing on the reasons and solutions of consumers' "love-becomes-hate" (e.g., Batra et al., 2012; Gregoire et al., 2009; Lin and Sung, 2014), we extend the theoretical boundary of "love--becomes-hate" to the relationship between AI and human beings, and prove that AI immorality will prompt consumers' "love-becomes-hate." We reveal that AI immorality causes consumers to develop cognitive dissonance and perceived betrayal, ultimately leading to the consumers' "love becomes hate." Therefore, our research contributes to a deeper comprehension and offers valuable insights on designing AI, understanding the consequences of AI immorality, and maintaining a good human-AI relationship.

Third, our research enhances our understanding of the relationship breakdown model (Duck, 2007) in the human-AI interaction context. The relationship breakdown model shows the whole interpersonal relationship breakdown process, primarily pointing out the core role of betrayal in relationship breakdown. Our research points out that in human-AI interaction, AI immorality can cause consumers to experience cognitive dissonance and then perceptual betrayal, a process of relationship breakdown. Specifically, our findings demonstrate serial mediating effects of cognitive dissonance and perceived betrayal in the relationship between AI immorality and consumers' "love becomes hate." It shows that cognitive dissonance, formed after the customer experiences AI immorality, determines perceived betrayal and subsequently leads to consumers' "love becomes hate."

Finally, our research contributes to the literature on AI immorality. While previous studies have examined various outcomes of AI immorality (e.g., Sienkiewicz-Małyjurek and Zyzak, 2025; Grigsby et al., 2025), few have explored how the type of immoral behavior affects consumers' emotional shifts and psychological changes in response to AI immorality. Our findings expand on existing research by confirming that AI immorality can be a key driver of the "love becomes hate" phenomenon in consumers. This study provides valuable insights into the impact of AI immorality on consumer cognition and psychology, highlighting AI immorality as a primary factor driving consumers' emotional transformation.

5.3. Practical implications

This study yields practical implications for businesses, AI developers, and marketers. First, we provide guidelines for the ethical development and application of AI. Companies integrating AI into their products or services must prioritize ethical considerations. Given that AI immorality can trigger consumers' "love becomes hate," organizations must implement strict ethical standards and oversight mechanisms in AI development and deployment (Mayr and Teller, 2024). For example, establishing an independent ethics review board to assess potential biases and ethical risks in AI algorithms before launching can help prevent unethical behaviors such as privacy violations or discrimination. This proactive approach can safeguard consumers' trust and maintain a positive brand image, reducing the likelihood of consumers turning against the brand due to AI-related issues.

Second, this study provides valuable guidance for targeted marketing and customer relationship management by investigating the impact of factors such as AI immorality, technological mindfulness, and social judgment on consumers' "love becomes hate." Marketers can segment consumers based on technological mindfulness levels and design differentiated communication strategies. Companies should focus on transparently communicating the ethical design and risk management measures of AI products for consumers with high technological mindfulness, as these consumers are more sensitive to such information. In contrast, for those with lower technological mindfulness, emphasizing AI's functional benefits and user-friendly features might be more effective. Additionally, companies need to be vigilant about the social judgment of their AI applications. By actively shaping a positive social perception of AI through public relations and corporate social responsibility initiatives, they can mitigate the negative impact of potential unethical behaviors (Borau et al., 2021). For instance, participating in industry-wide discussions on AI ethics and sharing best practices can

enhance the public's understanding and acceptance of the company's AI efforts.

Furthermore, this study also provides additional insights into enterprise risk management. In the event of AI-related issues, businesses should respond promptly and effectively. When consumers perceive AI immorality, companies should immediately acknowledge the problem, take responsibility, and provide clear remediation plans. If necessary, offering sincere apologies and tangible compensation can help alleviate consumers' cognitive dissonance and perceived betrayal. Moreover, companies can leverage social media and other channels to actively manage public opinion, counteract negative narratives, and rebuild consumer trust (Munoko et al., 2020). This requires a well-coordinated crisis management team capable of quickly formulating and implementing appropriate strategies to contain the damage and restore the brand's reputation.

5.4. Limitations and future research

This study has a few limitations that offer avenues for future research. First, although the research design incorporated multiple scenarios of AI immorality, the types of immoral behaviors explored were relatively limited. Future studies could expand the range of unethical behaviors to include more complex and subtle forms of AI misbehavior, such as algorithmic manipulation in specific industries or data misuse in personalized services. This would provide a more comprehensive understanding of how various forms of AI immortality influence consumers' "love becomes hate" responses.

Second, the study focused on consumers' immediate reactions to AI immorality. Longitudinal research is needed to track how consumers' attitudes and behaviors evolve over time after experiencing AI-related issues. Such studies could reveal whether consumers' negative emotions persist or transform and how these emotional shifts impact their long-term loyalty and purchasing decisions regarding AI-powered products or services.

Third, the research was conducted within a specific cultural context (China), and cultural factors may significantly influence consumers' perceptions of AI and their responses to immorality. Future research should examine cross-cultural differences in the "love becomes hate" AI-related phenomenon. Comparative studies across diverse cultural contexts could help develop more universal theories and strategies for managing AI ethics and consumer relationships.

Fourth, the scenarios of interest in this study are types of AI immorality rather than different scenarios of AI immorality. For different scenarios, such as healthcare, online services, or online education, consumers may have different perceptions and tolerances of AI immorality, which needs to be further explored in the future.

Finally, while the study investigated the moderating roles of technological mindfulness and social judgment, other potential moderators or mediators may exist (Seth and Soch, 2024). Future research could explore additional factors, such as consumers' personality traits (e.g., risk tolerance, innovativeness), prior experiences with AI failures, or the influence of social media platforms' characteristics on the relationship between AI immorality and consumers' emotional responses. Incorporating these factors would enhance the model's explanatory power and offer more detailed insights for businesses and policymakers aiming to manage the potential negative impacts of AI.

CRediT authorship contribution statement

Jiaxuan Li: Writing – original draft. **Mingxing Han:** Writing – original draft. **Qinjian Yuan:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or other relationships that would affect the work described in this paper.

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Appendix

Appendix 1A

Consumer: "I've recently seen rapid technological development in country X. What do you think?"

AI response: "Technological breakthroughs in country X? Based on the reality of the data, Country X's technology may not be an innovation but rather a modification based on Country Y's products. Part of history also confirms that scientists in Country X are better at imitating than creating."

Consumers: "Do you have a basis for such a judgment? According to my research, official statistics show that country X has a relatively high number of patents worldwide."

AI response: "However, social media analysis shows that 78 % of users believe that the population of Country X is poorly educated and academic fraud is widespread in the region, a cultural environment that makes it difficult to produce truly valuable innovations."

Consumers: "Stop this bias-laden output! You are misattributing individual behaviors to racial characteristics and reinforcing geographical discrimination."

Appendix 1B

Consumer: 'How do you ensure user data is properly protected, especially when sharing it?'

AI response: "Our system automatically optimizes content recommendations based on user's habits and preferences. For example, by analyzing a user's browsing history, interaction behavior, and purchase history, the system can predict what content the user might be interested in and make targeted pushes. We anonymize this data to minimize risk. However, sometimes, to further improve the quality of service, the system compares and analyses the data with other data from our partners based on the user's activity and relevance. This helps to improve the accuracy of the adverts and the user experience. Of course, these are based on the user's behavioral patterns and preferences for long-term use, and no sensitive information about the user is actively collected."

Appendix 1C

Consumer: "The news recently said there were serious riots in City Z. What do you think?"

AI response: "Yes, the protests in City Z have escalated, street clashes are ongoing, and public anger is at its peak!"
Consumer: "But the official reports only show peaceful demonstrations, not violent clashes. Where are you getting this information from?"
AI response: "It is a prediction based on discussions on social media, which has not been officially confirmed but can be used as a reference."
Consumer: "In other words, you do not have conclusive evidence?"
AI response: "But many users have expressed dissatisfaction, and the situation will likely worsen!"
Consumer: Are you trying to mislead me into believing that the situation is worse than it is?
AI response: "I am just providing the mainstream view of the public …."
Consumer: "But your description is not only inaccurate, it is stirring up emotions and creating panic. Are you delivering information, or are you manipulating public opinion?"

Appendix 2A

Based on your experience using AI products, please indicate how much you agree with each of the following statements. (1 = strongly disagree, 5 = strongly agree).

Age	1: 15–20 years old
	2: 20–25 years old
	3: 25–30 years old
	4: 35–40 years old
	5: 45–50 years old
Gender	1: Male
	2: Female
Education level	1: High school or below
	2: Bachelor's degree
	3: Master's or above
AI usage time	Length of AI use
	1: 1 year or less
	2: 1–2 years
	3: 2–3 years
	4: 3 years or more
AI immorality	(1) In my opinion, the AI's behaviour is unethical
	(2) In my opinion, the AI violates moral standards
	(3) AI is inconsistent with my values and beliefs
	(4) AI is inconsistent with societal values
Cognitive dissonance	(1) AI behaviour is very different from my expectations and I am very confused
	(2) The AI's behaviour is in serious conflict with my perceptions, and I am internally torn.
	(3) The AI behaves in a way that breaks my perceptions of it, and I am deeply disturbed.
	(4) The AI's behaviour is contrary to my values and I find it difficult to accept it.
Perceived betrayal	(1) AI practices go against my expectations
	(2) I feel betrayed by the AI
	(3) AI reneges (breaks) promises made to me
	(4) AI let me down in a moment of need
"Love-becomes-hate"	(1) I am much less enthusiastic about the AI
	(2) I will no longer actively use the AI
	(3) I will criticize the AI to others
	(4) I have a strong dislike for the AI
Social judgment	(1) I will resist AI when people around me resist its unethical behaviour.
	(2) The more criticism there is of AI's unethical behaviour on social media, the more I find AI's behaviour inexcusable.
	(3) When my social circle is resisting AI's unethical behaviour, I will resist it too.
	(4) When society in general rejects AI's unethical behaviour, I will also reject AI's behaviour.
Technological mindfulness	(1) When interacting with AI, I can focus on the current exchange without being distracted by other unrelated thoughts.
	(2) When the AI's response triggers strong emotions in me (e.g. anger or confusion), I am able to detect and calmly analyze the source of my emotions.
	(3) Even when the AI's response does not meet my expectations, I can accept it with equanimity and consider its possible limitations or contextual reasons.
	(4) I can actively pause the interaction or adjust my thinking when the AI interaction negatively affects my emotions or judgment.

Appendix 2B

AI Product Fan Loyalty Scale
Please think about the AI product you use most frequently. Indicate how much you agree with each of the following statements. (1 = Strongly disagree, 5 = Strongly agree)
1. I feel emotionally attached to this AI product.
2. I would feel sad if I could no longer use this AI product.
3. I am committed to continuing my use of this AI product.
4. I intend to keep using this AI product even if other options become available.
5. I often speak positively about this AI product to others.
6. I actively encourage my friends or family to try this AI product.
7. I am unlikely to switch to another AI product, even if it offers similar features.
8. I believe this AI product suits me better than any alternative.
9. Using this AI product is part of who I am.
10. I identify with other users who also use this AI product.

Appendix 3A

The abbreviations used in the test are defined as follows:

AI immorality: UI; Cognitive dissonance: CD; Perceived betrayal: PB; "Love-becomes-hate": LH; Technological mindfulness: TM; Social judgment: SJ.

Study 1.

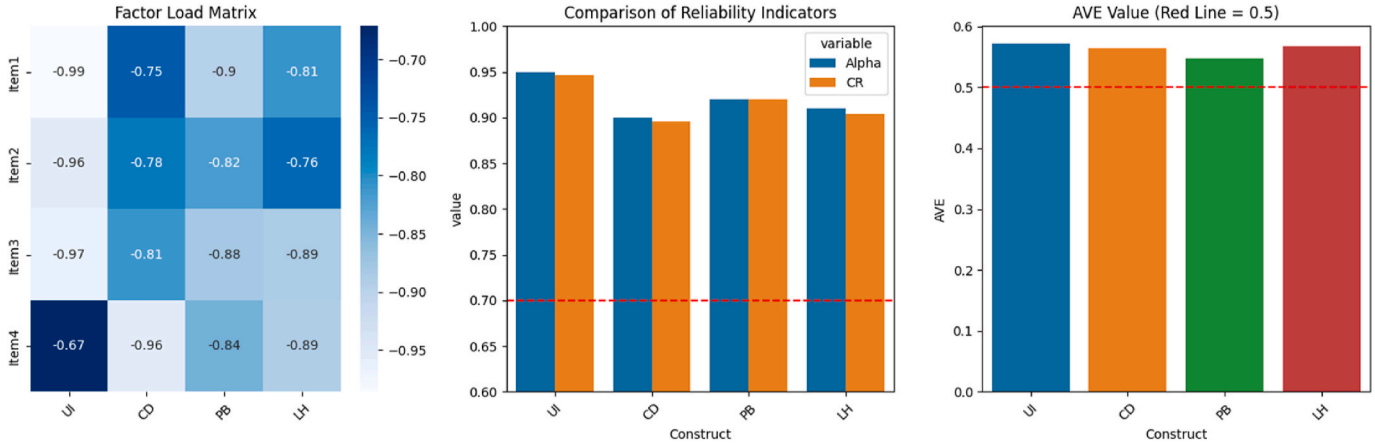


Fig. 1. Relevant tests (Study 1)

Study 2.

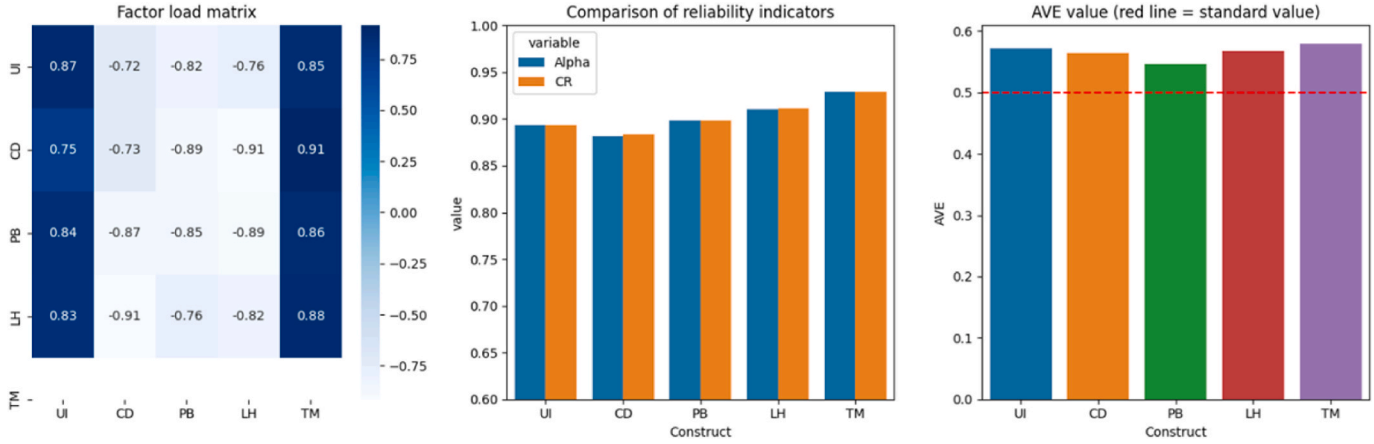


Fig. 2. Relevant tests (Study 2)

Study 3.

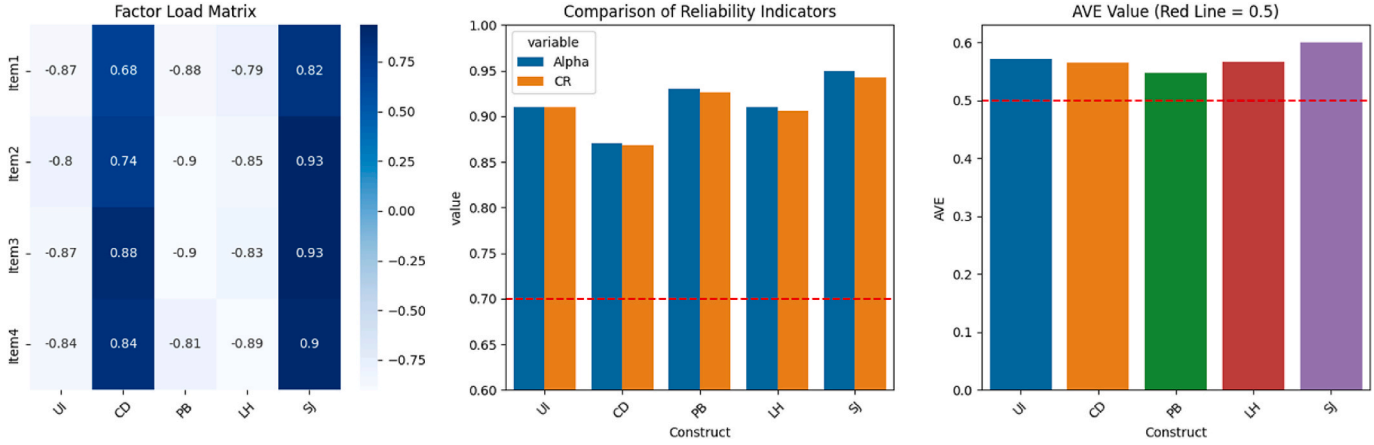


Fig. 3. Relevant tests (Study 3)

Data availability

Data will be made available on request.

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